Procurement Restructuring & Supplier Intelligence Dossier

Tender UP-1E88DAA1 - Re-tender: The Project proposes to procure a **Millet Processing and Value Addition Plant**, primarily comprising a **Primary Processing Line, Milling Line, Cold Extrusion Line, and Bakery Line**.
Generated 2026-11-13T04:41:32+00:00 by TenderShield Nexus (decision support - not an award decision)

Keep the procurement as one integrated contract: splitting does not improve procurement value under the stated weights.

A. Original procurement

Estimated value: unknown. Current structure: 1 package(s): Industrial & process machinery.

ID	Component	Items	Value	Value basis	Source pages
C1	Industrial & process machinery	9	unknown	unknown	1

B. Proposed restructuring

1 package(s): Industrial & process machinery

C. SME opportunity map

Pkg	Scope	Value	Turnover req.	Capable firms	Meet scale rule	SME candidates	Bidders (ind.)
P1	Industrial & process machinery	unknown	unknown	34	0	33	2.5

P1 potential supplier pool (top 6 of 34)

Supplier	Country	Status	MSME status	Score	Evidence
BATLIBOI	India	LIMITED_SCALE_EVIDENCE	SIZE_PROXY_SMALL	0.655	2 awards in Industrial & process machinery (largest [5.401])
PERFECT TRADING CORPORATION	India	LIMITED_SCALE_EVIDENCE	SIZE_PROXY_SMALL	0.655	2 awards in Industrial & process machinery (largest [8.783])
M/S PICKS MEDICAL SYSTEMS PVT. LTD., MUMBAI	India	LIMITED_SCALE_EVIDENCE	SIZE_PROXY_SMALL	0.655	2 awards in Industrial & process machinery (largest [3.707])
SHIV M/C TOOLS	India	LIMITED_SCALE_EVIDENCE	SIZE_PROXY_SMALL	0.655	2 awards in Industrial & process machinery (largest [4.809])
M/S. USHA DRAGER LTD.	India	LIMITED_SCALE_EVIDENCE	SIZE_PROXY_SMALL	0.655	2 awards in Industrial & process machinery (largest [2.156])
KENDRO LABORATORY PRODUCTS GMBH	Germany	LIMITED_SCALE_EVIDENCE	SIZE_PROXY_SMALL	0.615	1 awards in Industrial & process machinery (largest [6.158])

D. Supplier relationships (secondary objective)

10 shortlisted suppliers screened; 0 relationship(s) require review. Signals for human review only. Relationships are not evidence of wrongdoing.

Supplier A	Supplier B	Observed link	Review score	Ordinary explanations
BATLIBOI	PERFECT TRADING CORPORATION	same_project	0.1	Large projects legitimately award separate packages to different firms
BATLIBOI	SHIV M/C TOOLS	same_project	0.1	Large projects legitimately award separate packages to different firms
PERFECT TRADING CORPORATION	SHIV M/C TOOLS	same_project	0.1	Large projects legitimately award separate packages to different firms

E. Scenario comparison

Scenario	Packages	Score	Competition	Coverage	SME access	Coordination	Integration risk	Scale loss
A	1	0.35	0.5	1.0	0	0.0	0.0	0

Scenario	Packages	Score	Competition	Coverage	SME access	Coordination	Integration risk	Scale loss
B	1	0.35	0.5	1.0	0	0.0	0.0	0
C	1	0.35	0.5	1.0	0	0.0	0.0	0

Weight sensitivity: the recommended structure stays best in 100% of 300 runs with every weight perturbed (each weight x U(0.5, 1.5)). Weights: {"competition": 0.2, "coordination": 0.1, "coverage": 0.25, "integration_risk": 0.15, "scale_loss": 0.05, "sme_access": 0.25}

F. Recommendation

Decision support only. The procurement authority decides the structure, verifies supplier evidence and MSME status, and reviews any flagged relationships.

[WARNING] Component values unknown - Upload the BOQ, or enter the tender's estimated value

[INFO] MSME status shown as a size proxy unless Udyam evidence was imported - Import verified Udyam records via POST /api/market/udyam

Data sources and models

Market scope: ['India']; 7434 award records in scope from worldbank_awards:major_contract_awards:india, prozorro:ua.

capability_classifier: TF-IDF (word 1-2g + char 3-5g) + multinomial logistic regression, lexicon-blended at inference. Trained 2026-11-13T04:35:49+00:00 on Common Procurement Vocabulary (CPV 2008) (9454 rows); Annotated tender lines (53 rows). Metrics: {"cv_accuracy": 0.7875, "cv_folds": 5, "cv_macro_f1": 0.7383}

- CPV descriptions are short catalogue phrases; long Indian BOQ lines are a domain shift - the lexicon blend and the annotated-line evaluation exist for this reason.

- Single label per text; a mixed BOQ line is assigned its dominant capability.

competition_model: HistGradientBoostingRegressor (Poisson loss) on log10(value Rs Cr) + capability one-hot. Trained 2026-11-13T04:35:51+00:00 on Prozorro lots (Ukraine public procurement, 2019-2023) (92438 rows). Metrics: {"mae": 1.0558, "mae_baseline": 1.064, "mean_bidders_test": 1.979, "poisson_deviance": 0.6907, "poisson_deviance_baseline": 0.7008, "split": "time: train 2019-2021, test 2022+", "test_rows": 14170, "train_rows": 78268}

- Learned on Ukrainian open tenders; used as a relative indicator between scenarios, not a forecast of Indian bidder counts. Recalibrate with CPPP bid-opening data.

- Lots in the data carry one CPV division, so the model has no notion of bundling; the bundling penalty comes from real supplier co-capability evidence instead.

- Observational data: larger lots in this data attract slightly more bidders; that is a correlation (buyers, procedures and markets differ), not a causal effect of lot size.

Capability taxonomy keys: civil_works = Civil & building construction, electrical = Electrical systems & power, signalling_control = Signalling, traffic & safety control, instrumentation_sensors = Instrumentation, sensors & metering, ict_software = Software, ICT & control systems, telecom_networks = Telecom & communication networks, mechanical_hvac_plumbing = Mechanical, HVAC, pumps & plumbing, industrial_machinery = Industrial & process machinery, structural_fabrication = Structural steel & specialised fabrication, consultancy_design = Design, engineering & supervision services, operation_maintenance = Operation & maintenance services, vehicles_equipment = Vehicles & heavy equipment, medical_lab = Medical & laboratory equipment, other_goods_services = Other goods & services